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EDUCATION

B.Eng, Mechatronics Engineering

Ahmadu Bello University, Zaria

2020 – 2026 · Awaiting results

High School Diploma (SSCE)

Essence International School

2016 – 2019

CORE SKILLS

ROS 2 / ROS

Python, C / C++

PX4 & ArduPilot (SITL)

Gazebo, CARLA simulation

Embedded: Raspberry Pi, Orange Pi 5,

ESP32

Computer vision: MediaPipe, OpenCV

Edge AI: YOLOv8, ONNX, RKNN

Hardware bring-up & testing

Flight-log analysis (ArduPilot)

Linux (Ubuntu / Pop!_OS)

Git & GitHub

CAD: SolidWorks

CERTIFICATIONS

Scientific Computing with Python

freeCodeCamp · Jan 2024

CCNA: Introduction to Networks

Cisco · Nov 2021

Autonomous Programming

Competition

Shell Eco-marathon · May 2025

INTERESTS

UAV & autonomous systems

Robot locomotion

Edge AI for African problems

Rocket science & space tech

Robotics Engineer

Mechatronics Engineering | ROS 2 | Embedded Systems | Autonomous UAVs

ABOUT ME

I am a Mechatronics Engineering student at Ahmadu Bello University, Zaria, with my final-year defence and examinations complete and results pending, and a Robotics Engineer Intern who builds and debugs autonomous systems end to end, from ROS 2 nodes to the hardware they run on. I move comfortably between simulation (Gazebo, PX4/ArduPilot SITL, CARLA) and real hardware bring-up, testing and repair. My final year project is a hybrid VTOL aircraft for Sahel vegetation monitoring, where I traced a real lift-authority failure on the physical airframe and then validated the corrected propulsion analysis against the aircraft's own flight data. I care about building autonomous systems for African ecological and security problems, and I am driven by mastery over shortcuts, with a long-term goal of becoming a world-class roboticist.

RESEARCH

Bandwidth-Constrained Aerial Survey: Measured Uplink Capacity and Onboard Vegetation Indexing

Submitted to ICRA 2027

- Built on the hybrid VTOL platform below, using measured link capacity rather than assumed figures.

EXPERIENCE

Robotics Engineer Intern Jun 2025 – Present

Terrahaptix · Idu Industrial, Abuja

- Build ROS 2 perception and control pipelines for UAV autonomy, including a 3D perception stack for sensor fusion.
- Run PX4/ArduPilot SITL with Gazebo to validate flight behaviour before hardware testing.
- Bring up and test edge-AI inference on companion computers (Orange Pi 5 / NPU) alongside flight-stack tuning.
- Work across simulation and physical hardware within the same development cycle, on production timelines.

Prototyping Engineer (Remote, Volunteer) 2024 – 2025

FundusAI Ltd

- Contributed to the design, simulation and prototyping of Lumina, a handheld smartphone-linked fundus camera for diabetic retinopathy screening at the point of care.

Junior Mechanic (Part-time) Feb 2022 – Present

NorthMech 16 Autoworks

- Perform computer diagnostics on BMW vehicles and troubleshoot engine and system faults.
- Assist with engine removal, installation and rebuilds; hands-on mechanical and electrical fault-finding.

SELECTED PROJECTS

SahelSense — Hybrid VTOL QuadPlane for Sahel Vegetation Monitoring

Final Year Project · ABU Zaria · github.com/mk-bardi/Project-Timbuktu

- Designed, fabricated and flew a 1.9 m quad-lift plus pusher QuadPlane, built from scratch in foam and composite; **28 flights logged**, including autonomous waypoint missions with automatic VTOL transition.
- Diagnosed a hover failure through a controlled elimination sequence rather than gain tuning, isolating inadequate lift authority against a mis-sized propeller; resolved it with a motor and propeller change on the physical airframe.
- Validated the corrected propulsion analysis against flight data: the autopilot's converged hover throttle (**Q_M_THST_HOVER = 0.5496**, a thrust-to-weight of 1.82:1 at 5.5 kg) fell inside the density-corrected predicted band.

- Onboard software: GLI vegetation-index module and YOLOv8n inference via ONNX on a Raspberry Pi 3B, with MAVLink telemetry bridged into ArduPlane SITL.
- Stack: Python, ROS 2, ArduPilot on SpeedyBee F405 Wing, Gazebo, Raspberry Pi 3B, SIYI A8 Mini.

Vision-Controlled Robotic Hand & Forearm (InMoov)

Ongoing · github.com/mk-bardi/Vision-Based-Robotic-Hand-with-media-pipe

- Built a MediaPipe vision pipeline that mirrors live human hand gestures onto a 3D-printed InMoov hand and forearm, from the camera feed alone.
- Assembled tendon-driven actuation on MG996R servos via a PCA9685 driver, with an ESP32-CAM streaming live video for gesture capture.
- Ported the control pipeline from laptop inference to standalone embedded operation on a Raspberry Pi 4B / Orange Pi.

Autonomous Driving in CARLA + ROS 2 UGV Pipeline

Team Lead · Shell Eco-marathon Autonomous Programming Competition 2025 · github.com/mk-bardi/ABU_shell_auto

- Built a full ROS 2 perception-and-control pipeline for a tracked UGV, including a 3D perception stack for sensor-fusion experiments.
- Implemented A* route search over CARLA's road-network graph on the standard library, replacing a networkx dependency, running live on every replan.
- Precomputed the goal visiting order as a 2-opt TSP tour over 14 goals (1795 m against 1975 m for nearest-neighbour), baked into the waypoint file.
- Ran the traffic manager in synchronous mode with traffic lights held green, so timed 120 s runs of the same controller stay comparable to one another.

MEDIC-AID — Medication Dispenser with Caregiver Alerts & IoT Dashboard

Group Project · ABU Zaria · Designed and simulated, hardware in progress

- Designed a three-hopper dispenser that assembles a dose on a schedule and counts pills as piezoelectric sensors detect them falling, so an empty hopper or a jam cannot pass as a successful dose.
- Implemented fault classification that distinguishes an empty bottle from a suspected jam, and a Telegram command and event path for caregiver alerting.
- Hardware design around an ESP32-S2-WROVER, DS3231M RTC and three DRV8833 stepper drivers; full firmware verified in Wokwi simulation before board bring-up.

LEADERSHIP & TECHNICAL ROLES

- Team Lead, Shell Eco-marathon Autonomous Programming Competition (2025), Team ABU Zaria.
- Technical Lead, Shell Eco-Marathon Electric Category (2024/2025), Team ABU Zaria.
- Program Director, ABU Zaria Raspberry Pi Community (2024 – Present).
- Asst. Secretary General, NiMechE (2024); Social Media Manager, SPE (2024).